

Proofs of eleven congruence conjectures of Hanna from the OEIS, and the resolution of three more*

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September 2026

Abstract

Many entries of the On-Line Encyclopedia of Integer Sequences contributed by Paul D. Hanna define a power series by a functional equation and conjecture a congruence for its coefficients. We prove eleven such conjectures:

- the modulo-three conjecture of A389472;
- the parity conjectures of A120566, A240998, A273958, A295762, A301933, A338633, A338634 and A388734;
- the modulo-three conjectures of A274479 and A377100.

We also resolve three more:

- for A184894 and A107099 we prove the conjectured vanishing modulo 3 outside the exceptional indices. We also show that the stronger reading, non-vanishing at every exceptional index, is false. It fails at the OEIS index 365 of A184894, and at the coefficient of x^{729} (OEIS index 364) of A107099;
- for A361047 we prove the intended statement, which the entry states with an index shift.

Several of these results are quick consequences of classical results, which we point out. Modulo 2, the series of A240998, A295762, A301933, A338633 and A338634 reduce, up to an additive constant, to the Catalan series, and that of A273958, multiplied by x , to a composition of two Catalan series, so these parity patterns follow from the classical parity of the Catalan numbers. The results for A184894 follow quickly from the classical correspondence for linearised polynomials over $\mathbb{F}_3$. For A120566, A274479 and A388734, identities stated in the entries themselves give short proofs. For A107099 and A361047 the reduced series are classical series, identified by a routine uniqueness argument. Only for the modulo-three conjecture of A389472 and for A377100 did we find neither an earlier proof nor a classical derivation. The proofs reduce the functional equations modulo a prime and use the Frobenius identity, formal derivatives, explicit solutions with an x -adic uniqueness argument, and Lucas's theorem. Every theorem is stated for every integer power series satisfying the defining equation, and every theorem is formally verified in Lean 4. The proofs were found by AI coding agents working under the author's direction; their role is described at the end of the paper.

* *Correction.* This version corrects version 1 (released 2026-09-26), which did not cite the connections with classical results and with identities stated in the OEIS entries described in Section 1 under “Relation to classical results” and in Sections 2, 5 and 6. Several of the conjectures proved here are quick consequences of classical results, whereas version 1 implied that all the proofs were new. The theorems, their proofs and the Lean formalisation are unchanged.

1 Introduction

Throughout, $A(x) = \sum_{n \geq 0} a(n)x^n \in \mathbb{Z}[[x]]$, A' is its formal derivative, and for a series h with $h(0) = 0$ we write $A(h)$ or $A \circ h$ for the composition $\sum_n a(n)h^n$. The results are summarised in Table 1; each conjecture is stated in the corresponding entry [4]. The theorems hold for every integer solution of the defining equation, together with the normalisation of the first terms fixed by the entry where the equation alone does not determine it. The equation and these initial terms determine the sequence by a triangular recursion with integer solutions. There, the new coefficient appears with coefficient 1 or 2. Where it appears with 2, the required divisibility holds: for A274479 by Lemma 1 modulo 2, and for A107099 by writing $A = x + 2B$. For the continued fractions, the tails exist as x -adic limits. Hence the theorems apply to the sequences in the OEIS. These existence arguments are not formalised; the Lean theorems hold for every integer solution.

Entry	Defining equation	Result
A389472	$A(x^2 + x^3) = x^2(1 + A)$	$3 \mid a(3n - 1), n > 1$
A240998	$A^2 = x + A(x + 2x^2)$	$a(n) \text{ odd} \iff n = 2^k \ (n \geq 1)$
A295762	$A(x - 2A(x^2)) = x + A(x^2)$	$a(n) \text{ odd} \iff n = 2^k \ (n \geq 1)$
A273958	$xA + x^2A^2 = C^2, C = x + C^2$	$a(n) \text{ odd} \iff n = 2 \cdot 4^k - 1$
A301933	$A = x(1 + 4AA')/(1 + AA')$	$a(n) \text{ odd} \iff n = 2^k$
A338633	$1 = A - x/(A - 2^3x/(A - 3^3x/\cdots))$	$a(n) \text{ odd} \iff n = 2^k \ (n \geq 1)$
A338634	$1 = A - x/(A - 2^4x/(A - 3^4x/\cdots))$	$a(n) \text{ odd} \iff n = 2^k \ (n \geq 1)$
A388734	$A = 1 + xA^2 + x^2(1 - x)A^3$	all $a(n)$ odd
A120566	$A = A \circ A - xA \circ A \circ A$	all $a(n)$ odd ($n \geq 1$)
A377100	$A = A(x^3)/A(x^2) + A^2$	$a(n) \equiv 1 \pmod{3} \ (n \geq 1)$
A274479	$A^2 = A(x^2/(1 - 2x - 4x^2))$	$a(n) \equiv 1 \pmod{3} \ (n \geq 1)$
A361047	$A(x - x^3A'^2) = x$	$[x^m]A \equiv [m = 3^k] \pmod{3}$
A184894	iterates of $x + x^3$	vanishing proved; non-vanishing fails at index 365
A107099	$A \circ A = x + 4x^3$	vanishing proved; non-vanishing fails at x^{729}

Table 1: Summary. Normalisations and precise statements are given in the theorems below. Several of these results are quick consequences of classical results or of identities stated in the entries; see “Relation to classical results” below.

Prior work. The modulo-two conjecture of A389472 was proved earlier by Perez Fontelles [6] and, independently, in the `trureturing` repository [11]; both leave the modulo-three conjecture open.

Relation to classical results. Several of the conjectures proved here are quick consequences of classical results. Write $C_n = \binom{2n}{n}/(n+1)$ for the Catalan numbers and $C = \sum_{n \geq 1} C_{n-1}x^n$ for the Catalan series, so that $C = x + C^2$. Modulo 2, the series of A295762 and A301933, and those of A240998, A338633 and A338634 minus 1, all equal C ; this is how the proof of Theorem 5 proceeds. For A301933 it can also be read off from the remark in the entry that $C = x(1 + 2CC')/(1 + CC')$: since $2 \equiv 4 \equiv 0 \pmod{2}$, both equations reduce to $F = x/(1 + FF')$, which determines F . Hence $a(n) \equiv C_{n-1} \pmod{2}$ for $n \geq 1$, and these five patterns follow from the classical fact that C_n is odd if and only if $n = 2^k - 1$, a corollary of $\nu_2(C_n) = s_2(n+1) - 1$, where ν_2 is the 2-adic valuation and s_2 the binary digit sum [1, Theorem 2.1 in arXiv:math/0407326v1]. Lemma 3 is this fact. The continued fractions of A338633 and A338634 are the cases $e = 3, 4$ of a family whose cases $e = 1, 2$ are A000699 and A158119. All of them reduce to $A(A-1) = x$ modulo 2, and the parity of A000699 is recorded in A036987 (J. M. Campbell, 2016). For A273958, formula (1) of the entry, which solves the

defining equation as a quadratic in xA , gives $xA = -C(-C^2)$, hence $xA \equiv C(C(x^2)) \pmod{2}$. The congruence $C(C(x)) \equiv \sum_{k \geq 0} x^{4^k} \pmod{2}$ is also the key step of the proof of Hanna's conjecture for A374568 in the `trureturing` repository [10]. Parts (1) and (2) of Theorem 12 are an instance of the classical correspondence for linearised polynomials over $\mathbb{F}_3$, and the residues at the exceptional indices of A184894 and A107099 are, up to sign, Catalan numbers modulo 3; see the remark at the end of Section 6. For A361047 the reduced series coincides with that of the ternary numbers A001764, whose residues modulo 3 are known; see Remark 11. For A120566, A274479 and A388734 we found no classical connection, but identities stated in the entries themselves give short proofs; see the remarks after Theorems 6, 8 and 9.

Apart from the connections with classical results noted above, we found no earlier proof of any of the results in this paper in the OEIS entries and their histories, in the literature, or in the public repositories of proofs of OEIS conjectures that we searched (among them [7, 12]). For the modulo-three conjecture of A389472 and for A377100 we also found no classical connection. This reports the coverage of our searches, not a guarantee of priority.

2 Tools

Reduction of the coefficients modulo a prime p is a ring homomorphism $\mathbb{Z}[[x]] \rightarrow \mathbb{F}_p[[x]]$; it commutes with formal differentiation and with composition by series with zero constant term. We write $F = \overline{A}$ for the reduction and $f(n)$ for its coefficients.

Lemma 1 (Frobenius). *For $F \in \mathbb{F}_p[[x]]$ we have $F(x)^p = F(x^p)$.*

Proof. The map $G \mapsto G^p$ is a continuous ring endomorphism of $\mathbb{F}_p[[x]]$ that fixes $\mathbb{F}_p$ and sends x to x^p . $\square$

Lemma 2. *Let K be a field, let $h \in xK[[x]]$ have order at least two, and let $H \in K[[x]]$ satisfy $H(h) = H$. Then H is constant.*

Proof. Put $B = H - H(0)$, so $B(h) = B$. If $B \neq 0$, its order d is positive, while $\text{ord } B(h) \geq 2d > d$. $\square$

Lemma 3 (dyadic recursion). *Let $F \in \mathbb{F}_2[[x]]$ satisfy $F(0) = 0$ and $F = x + F^2$. Then for $n \geq 1$, $f(n) = 1$ if and only if n is a power of 2.*

Proof. By Lemma 1, $F = x + F(x^2)$, so $f(n) = [n = 1] + [2 \mid n]f(n/2)$ for $n \geq 1$. By strong induction, $f(1) = 1$, $f(n) = 0$ for odd $n \geq 3$, and $f(2m) = f(m)$. $\square$

Since the reduction of the Catalan series C satisfies $F = x + F^2$, Lemma 3 is the classical fact that C_n is odd if and only if $n = 2^k - 1$; see Section 1.

We also use the following observation repeatedly: if $D \in \mathbb{F}_p[[x]]$, $x^{n_0} \mid D$, and $x^n \mid D$ implies $x^{n+1} \mid D$ for all $n \geq n_0$, then $D = 0$.

3 The modulo-three conjecture of A389472

Theorem 4. *If $A(x^2 + x^3) = x^2(1 + A(x))$, then $a(3n - 1) \equiv 0 \pmod{3}$ for every $n > 1$.*

Proof. Let $F = \overline{A} \in \mathbb{F}_3[[x]]$ and $h = x^2 + x^3$. Then $F(h) = x^2(1 + F)$ and $h' = 2x + 3x^2 = -x$. Differentiating, $-xF'(h) = -x(1 + F) + x^2F'$, so $F'(h) = 1 + F - xF'$ after cancelling $-x$. Differentiating again, the terms F' cancel and $-xF''(h) = -xF''$, so $F''(h) = F''$. By Lemma 2, F'' is constant. For $n > 1$ the coefficient of x^{3n-3} in F'' is therefore zero, and it equals $(3n - 1)(3n - 2)f(3n - 1) = 2f(3n - 1)$. $\square$

4 Parity via the dyadic recursion

Theorem 5. *In each of the following cases, $a(n)$ is odd for $n \geq 1$ if and only if n is a power of 2:*

1. (A240998) $A^2 = x + A(x + 2x^2)$;
2. (A295762) $a(0) = 0$ and $A(x - 2A(x^2)) = x + A(x^2)$;
3. (A301933) $a(0) = 0$ and $A(1 + AA') = x(1 + 4AA')$;
4. (A338633, A338634) A satisfies the continued fraction $1 = A - x/(A - 2^e x/(A - 3^e x/(A - \dots)))$ with $e = 3$, resp. $e = 4$, in the following sense: there are series $T_0, T_1, \dots$ with constant terms ± 1 such that $(A - 1)T_0 = x$ and $(A - T_k)T_{k+1} = (k + 2)^e x$ for all $k \geq 0$.

Moreover (A273958), if $C = x + C^2$ and $xA + x^2A^2 = C^2$, then $a(n)$ is odd if and only if $n = 2 \cdot 4^k - 1$ for some $k \geq 0$.

Proof. (1) Modulo 2, $(x + 2x^2)^k \equiv x^k$, so $F^2 = x + F$. Then $G = F - f(0)$ satisfies $G(0) = 0$ and $G^2 = F^2 - f(0) = x + G$, and Lemma 3 applies to G . (2) Modulo 2 the inner series reduces to x , so $F = x + F(x^2) = x + F^2$. (3) Modulo 2 the equation reads $F + F^2F' = x$. In characteristic 2, $F'' = 0$ because $n(n-1)$ is even. Differentiating gives $F' = 1$, hence $F = x + F^2$. (4) Modulo 2, $(A - T_0)T_1 = 0$ with T_1 a unit, so $T_0 = A$ and $(A - 1)A = x$. Then $B = A - 1$ satisfies $B(0) = 0$ and $B + B^2 = x$, so $B = x + B^2$, and $a(n) = b(n)$ for $n \geq 1$.

For A273958, constant terms give $C(0)^2 = 0$. By Lemma 3, $c = \overline{C}$ is the indicator of the powers of 2. Put $B = x\overline{A}$, so that $b(n+1) = f(n)$ and $B + B^2 = c^2$, i.e. $B(x) + B(x^2) = c(x^2)$. Hence $b(n) = 0$ for odd n and $b(2m) = b(m) + c(m)$. The powers of 2 split into the disjoint sets $\{4^j\}$ and $\{2 \cdot 4^k\}$, and strong induction gives $b(n) = 1$ exactly for $n = 2 \cdot 4^k$. $\square$

5 Explicit solutions modulo a prime

In the next results we exhibit the reduction F explicitly and show that it is the only solution by the x -adic observation of Section 2.

Theorem 6 (A388734). *If $A = 1 + xA^2 + x^2(1 - x)A^3$, then every $a(n)$ is odd.*

Proof. Modulo 2, $U = 1/(1 - x)$ is a solution: $1 + (x + x^2)/(1 - x)^2 = 1/(1 - x)$ because $x + x^2 \equiv x(1 - x)$. Subtracting the equations for F and U gives $F - U = (F - U)x[F + U + x(1 - x)(F^2 + FU + U^2)]$, so $x^n \mid F - U$ implies $x^{n+1} \mid F - U$. Hence $F = U$. $\square$

Remark. Formula (4) of the entry, read as a formula for xA as in the program of the entry, gives another short proof: xA is the series reversion of $\varphi = x(1 - x - x^2)/(1 - x^3)$. (This holds because $B = (1 - x^3)/(1 - x - x^2)$ satisfies $B = 1 + \varphi B^2 + \varphi^2(1 - \varphi)B^3$ with $\varphi = x/B$; since φ has no constant term, the equation $Y = 1 + \varphi Y^2 + \varphi^2(1 - \varphi)Y^3$ has a unique power-series solution, so $A(\varphi) = B$, which is formula (5) of the entry.) Since $1 - x^3 = (1 - x)(1 + x + x^2)$, we have $\varphi \equiv x/(1 + x) \pmod{2}$, whose reversion is $x/(1 - x)$; hence $A \equiv 1/(1 - x)$.

Theorem 7 (A377100). *If $a(0) = 0$, $a(1) = 1$ and $A = A(x^3)/A(x^2) + A^2$, then $a(n) \equiv 1 \pmod{3}$ for $n \geq 1$.*

Proof. Modulo 3 the equation reads $F(x^2)(F - F^2) = F(x^3)$. The series $E = x/(1 - x)$ is a solution: both sides equal $x^3/(1 - x)^3$, because $(1 - x)^3 = 1 - x^3$ and $1 - 2x \equiv 1 + x$. For $D = F - E$ one computes $E(x^2)D(1 - F - E) = D(x^3) - D(x^2)(F - F^2)$. If $x^n \mid D$ with $n \geq 2$, the right-hand side is divisible by x^{2n+1} ; since $E(x^2)$ is x^2 times a unit and $1 - F - E$ is a unit, $x^{2n-1} \mid D$. As $x^2 \mid D$, we get $F = E$. $\square$

Theorem 8 (A274479). *If $a(0) = 0$, $a(1) = 1$ and $A^2 = A(x^2/(1 - 2x - 4x^2))$, then $a(n) \equiv 1 \pmod{3}$ for $n \geq 1$.*

Proof. Modulo 3, $h = x^2/(1 + x - x^2)$ and $E = x/(1 - x)$ satisfies $E(h) = x^2/(1 + x - 2x^2) = x^2/(1 - x)^2 = E^2$. For $D = F - E$ we have $D(F + E) = D(h)$. If $x^n \mid D$ then $x^{2n} \mid D(h)$, and $F + E = xW$ with $W(0) = 2$ a unit, so $x^{2n-1} \mid D$. As $x^2 \mid D$, we get $F = E$. $\square$

Remark. The entry already notes, in its comment (1), that $x/(1 - x)$ satisfies $F^2 = F(x^2/(1 - 2x + 2x^2))$. As $-4 \equiv 2 \pmod{3}$, this is the reduced equation, so with this identity the conjecture has a short proof: only the uniqueness argument above is needed.

Theorem 9 (A120566). *If $a(0) = 0$, $a(1) = 1$ and $A = A \circ A - x A \circ A \circ A$, then $a(n)$ is odd for every $n \geq 1$.*

Proof. Modulo 2, $E = x/(1 - x)$ satisfies $E \circ E = x/(1 - 2x) = x$, so $E \circ E - x E \circ E \circ E = x - x E = E$. Let $D = F - E$ and suppose $x^n \mid D$ with $n \geq 2$. Since composition preserves congruences modulo x^n and $F = x + O(x^2)$, we have $D \circ F \equiv D$, $E \circ F - E \circ E = D/((1 - F)(1 - E)) \equiv D$, and $x(F \circ F - E \circ E) \equiv 0$, all modulo x^{n+1} . Since $F \circ F - E \circ E = D \circ F + (E \circ F - E \circ E)$, subtracting the equations for F and E gives $D \equiv 2D \equiv 0 \pmod{x^{n+1}}$. As $x^2 \mid D$, we get $F = E$. $\square$

Remark. Formula (2) of the entry, $A(-A(-x)) = x$, gives a shorter proof. It follows from the defining equation: both $-A(-x)$ and the compositional inverse of A satisfy $B = B \circ B + x B \circ B \circ B$, and this equation has only one solution $B = x + O(x^2)$. Modulo 2 we have $-A(-x) \equiv A$, so formula (2) gives $F \circ F = x$, and the defining equation becomes $F = x - xF$, that is, $F = x/(1 + x)$.

Theorem 10 (A361047). *If $A(0) = 0$ and $A(x - x^3 A'^2) = x$, then $[x^m]A \equiv 1 \pmod{3}$ when m is a power of 3, and $[x^m]A \equiv 0 \pmod{3}$ otherwise.*

Proof. Modulo 3 let $S = \sum_{k \geq 0} x^{3^k}$. Then $S' = 1$ and, by Lemma 1, $S - S^3 = S - S(x^3) = x$, so $g \circ S = x$ for $g = x - x^3$. Hence $S \circ g = x$, i.e. $S(x - x^3 S'^2) = x$. For $D = F - S$ with $x^n \mid D$, the inner series $g_F = x - x^3 F'^2$ and $g_S = x - x^3 S'^2$ differ by $x^3 D'(F' + S')$, which is divisible by x^{n+2} . Therefore $0 = F(g_F) - S(g_S) = D(g_F) + (S(g_F) - S(g_S)) \equiv D \pmod{x^{n+1}}$. Hence $F = S$. $\square$

Remark 11. The entry A361047 writes $A = \sum_{n \geq 1} a(n)x^{2n-1}$ and conjectures $a(n) \equiv 1 \pmod{3}$ exactly for $n = (3^k - 1)/2$. With this indexing the statement is contradicted by the data ($a(2) = 1$). It is correct for the index $i = n - 1$, since x^{2i+1} with $i = (3^k - 1)/2$ is x^{3^k} . Theorem 10 is this intended statement.

Over $\mathbb{Z}$ the reversion of $x - x^3$ is $V = \sum_{k \geq 0} t(k)x^{2k+1}$, where $t(k) = \binom{3^k}{k}/(2k+1)$ are the ternary numbers (A001764), since $V - V^3 = x$. Modulo 3 this reversion is S , so Theorem 10 says that $a(n) \equiv t(n-1) \pmod{3}$. The residues of $t(k)$ modulo 3 form A113047, which is 1 for $k = (3^j - 1)/2$ and 0 otherwise; this was conjectured by P. Barry and proved by K. Ryde in 2021 [9]. The statement in A361047 is this pattern without the index shift. The remaining step, $F = S$, or equivalently $F' = 1$, which reduces the defining equation modulo 3 to $F(x - x^3) = x$, follows by the routine uniqueness argument in the proof above.

6 Additive series modulo three

Call $G \in \mathbb{F}_3[[x]]$ *additive* if $[x^m]G = 0$ whenever m is not a power of 3. By Lemma 1, $G^{3^j} = G(x^{3^j})$, so the composition of two additive series is additive.

Theorem 12 (A184894). *Let $P_0 = x$, $P_{m+1} = P_m + P_m^3$, and $a(m) = [x^{2m-1}]P_m$ for $m \geq 1$.*

1. If $2m - 1$ is not a power of 3, then $3 \mid a(m)$.
2. If $2m - 1 = 3^j$, then $a(m) \equiv \binom{m}{j} \pmod{3}$.
3. $3 \mid a(365)$, although $2 \cdot 365 - 1 = 3^6$.

Proof. Modulo 3, $P_{m+1} = P_m + P_m(x^3)$, so the coefficients $c(m, k) = [x^k]P_m$ satisfy $c(m+1, k) = c(m, k) + [3 \mid k]c(m, k/3)$ with $c(0, k) = [k = 1]$. By induction, $c(m, k) = 0$ unless k is a power of 3, and $c(m, 3^j) = \binom{m}{j}$. Finally $\binom{365}{6} \equiv 0 \pmod{3}$ by Lucas's theorem, since $365 = (111112)_3$ and $6 = (20)_3$. $\square$

The entry conjectures that $a(m) \equiv 0 \pmod{3}$ except at $m = (3^n + 1)/2$. Part (1) proves this statement as written, namely vanishing outside the exceptional indices. Part (3) shows that the stronger reading, that $a(m)$ is nonzero modulo 3 at every exceptional index, is false; the first failure is at $n = 6$. By Lucas's theorem, $a((3^n + 1)/2) \not\equiv 0$ exactly when every base-3 digit of n except the last is at most 1.

Theorem 13 (A107099). *Let $A(0) = 0$, $[x^1]A = 1$ and $A \circ A = x + 4x^3$. Then $3 \mid [x^n]A$ whenever n is not a power of 3, and $3 \mid [x^{729}]A$.*

Proof. Modulo 3 the equation reads $F \circ F = x + x^3$. Suppose m is not a power of 3 and $f(j) = 0$ for all non-powers $j < m$. Let T be the truncation of F below degree m ; it is additive, so $[x^m]T \circ T = 0$. Since $F = T + f(m)x^m + O(x^{m+1})$ and $f(1) = 1$, we get $[x^m]F \circ F = 2f(m)$, while $[x^m](x + x^3) = 0$; hence $f(m) = 0$. Thus $F = \sum_k c_k x^{3^k}$ is additive, and comparing coefficients of x^{3^k} in $F \circ F = \sum_i c_i F(x^{3^i})$ gives $c_0 = 1$ and $2c_k + \sum_{0 < i < k} c_i c_{k-i} = [k = 1]$ for $k \geq 1$. This yields $(c_1, \dots, c_6) = (2, 1, 1, 2, 1, 0)$. $\square$

The entry writes $A = \sum_{n \geq 0} a(n)x^{2n+1}$ and conjectures that $[x^n]A \equiv 0 \pmod{3}$ except at $n = 3^k$. Theorem 13 proves this statement as written. It also shows that the stronger reading, non-vanishing at every power of 3, is false: $[x^{729}]A = a(364) \equiv 0 \pmod{3}$. The OEIS b-file of A107099 already shows this zero.

Remark. Parts (1) and (2) of Theorem 12 are an instance of the classical correspondence between linearised polynomials over $\mathbb{F}_3$ and their conventional 3-associates, $\sum_j c_j x^{3^j} \leftrightarrow \sum_j c_j t^j$, under which composition corresponds to multiplication [5], [2, Lemma 3.59]: since $x + x^3$ corresponds to $1+t$, we get $P_m \equiv \sum_j \binom{m}{j} x^{3^j} \pmod{3}$. Likewise, the correspondence, which extends to series, shows that the additive series $\sum_j \binom{1/2}{j} x^{3^j}$, which corresponds to $(1+t)^{1/2}$, composes with itself to $x + x^3$ modulo 3. For $m \geq 2$ the coefficient of x^m in $F \circ F$ is $2f(m)$ plus terms in lower coefficients, so by a routine uniqueness argument the reduction F in Theorem 13 is this series. For $j \geq 1$ we have $\binom{1/2}{j} = 2(-1)^{j-1}C_{j-1}/4^j \equiv (-1)^j C_{j-1} \pmod{3}$, and $\binom{(3^j+1)/2}{j} \equiv \binom{1/2}{j} \pmod{3}$ by Lucas's theorem, because $(3^j + 1)/2 \equiv 1/2 \pmod{3^j}$ and $j < 3^j$. So in both entries the residue at an exceptional index with $j \geq 1$ is $(-1)^j C_{j-1}$ modulo 3. By [1, Theorem 5.2 in arXiv:math/0407326v1], it is nonzero exactly when every base-3 digit of j except the last is 0 or 1. Both failures come from $3 \mid C_5 = 42$.

7 Verification

Numerical checks. For each entry, one of two checks was made. Either a prefix of the sequence was recomputed from the defining equation and compared with the b-file of the entry, or the b-file was substituted into the defining equation and checked modulo two primes. The lengths and moduli vary between entries and are recorded with the verification programs [8]. All checks agree with the theorems. They are cross-checks only; the theorems rest on the proofs and on the Lean verification.

Formal verification. All theorems are verified in Lean 4 (version 4.33.1, Mathlib [3] commit 0df444a3) for every integer power series satisfying the stated hypotheses. The Lean files are 0eis389472Mod3 (Theorem 4) and, for the other entries, HannaAnnnnnnn with the A-number of the entry (the file HannaA338633 treats both continued fractions). A separate check confirms that `A.subst h` denotes the composition $A(h)$. For example, Theorem 5 for A273958 reads, in ASCII notation,

```
theorem hanna_a273958 (C A : PowerSeries Int) (hC : C = X + C ^ 2)
  (hA : X * A + X ^ 2 * A ^ 2 = C ^ 2) (n : Nat) :
  Odd (coeff n A) <-> exists k, n + 1 = 2 * 4 ^ k
```

The project was rebuilt from a clean directory. The recorded axiom audits of the main results (`#print axioms`) report only the standard axioms `propext`, `Classical.choice` and `Quot.sound`; no native evaluation is used. The code and logs are available at [8].

Tool and computational resource disclosure

This work was carried out between 25 and 26 September 2026 in a workflow directed by the author, using two AI coding agents: OpenAI Codex (desktop application; models `gpt-6-astra` and `gpt-6-sol`) and Anthropic Claude Code (model Claude Opus 5.5). The author chose the research programme and directed the agents. Codex found and formalised the proof of Theorem 4. Claude Code agents searched the OEIS for candidate conjectures, found and formalised the other proofs, and drafted this text. Each proof was checked by a second agent, and the agents searched the literature for previous proofs. After version 1 was released, a further agent check, which compared the reduced series with classical sequences, found the connections with classical results and with identities stated in the OEIS entries described in Section 1; Claude Code drafted the corresponding changes, which an independent Claude agent checked. The author is not a professional mathematician. The author has read version 1 of the paper and, using a side-by-side table, compared the statements of the Lean theorems with the theorems stated here, which are unchanged in this version. The correctness of the results rests on the written proofs and on the Lean formalisation; the numerical checks described in the previous section are additional cross-checks. In addition, each proof was checked by an AI agent other than the one that found it, and the formal statements were compared with the OEIS entries by a second agent. AI systems are not authors of this paper; the author takes full responsibility for its content.

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